

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: B.Tech. Branch : Computer Engineering/Computer Science and Engineering****Subject Code & Name: Database Systems(BTCOC501)****Semester : V****Max. Marks: 60****Date: 06/02/2025****Duration: 3 Hr.****Instructions to the Students:**

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which type of data can be stored in the database? a. Image oriented data b. Text, files containing data c. Data in the form of audio or video d. All of the above		1
2	Database languages can be used to _____ the data in the database. a. Read b. Store c. Update d. All of the Mentioned.		1
3	A database is _____ with Data Manipulation Language. a. Accessed and Multiplied b. Added and Manipulated c. Access and Manipulated d. Added and Multiplied		1
4	What is a relation in RDBMS? a. Key b. Table c. Row d. Data Types		1
5	The function that an entity plays in a relationship is called that entity's _____. a. Participation b. Position c. Role d. Instance		1
6	The descriptive property possessed by each entity set is _____. a. Entity b. Attribute c. Relation d. Model		1

7	Database _____ which is the logical design of the database, and the database _____ which is a snapshot of the data in the database at a given instant in time.		1
	a. Instance, Schema b. Relation, Schema c. Relation, Domain d. Schema, Instance		
8	The intersection operator is used to get the _____ tuples.		1
	a. Different b. Common c. All d. Repeating		
9	The number of attributes in relation is called as its _____		1
	a. Cardinality b. Degree c. Tuples d. Entity		
10	A unit of storage that can store one or more records in a hash file organization is denoted as _____		1
	a. Buckets b. Disk pages c. Blocks d. Nodes		
11	A hash function must meet _____ criteria.		1
	a. Two b. Three c. Four d. None of the Mentioned		
12	Which of the following makes the transaction permanent in the database?		1
	a. View b. Commit c. Rollback d. Flashback		
Q. 2	Solve the following.		12
A)	Define an Entity and Attribute. Explain the different types of attributes that occur in ER diagram model with an example.		6
B)	Draw the symbols of ER diagram. Draw a diagram for library management system		6
Q.3	Solve the following.		12
A)	Define and differentiate the following relational algebra operators with suitable example 1. Cartesian product 2. Natural join		6
B)	What is relational algebra? Explain basic operations along with symbols of relational algebra.		6
Q. 4	Solve Any Two of the following.		12
A)	What is view what are its advantages explain views in SQL with example.		6

B)	What is normalization? Why we need to normalize the database tables.		6
C)	Write the SQL syntax for the following with the example 1. Select 2. Alter 3. Update		6
Q.5	Solve Any Two of the following.		12
A)	Explain 1 st , 2 nd and 3 rd normal form with the help of suitable example		6
B)	What are the features of good relational Designs?		6
C)	Explain The Static Hashing and Dynamic Hashing.		6
Q. 6	Solve Any Two of the following.		12
A)	Define transaction and their properties with suitable example		6
B)	Explain ACID properties in details.		6
C)	Explain Time Stamp-Based Protocols.		6
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